SalesGenie AI | AI/ML Engineer Handoff

Exact files, honest commits, Windows CMD steps and merge flow | Checked 29 October 2026

Answer to the team workflow: Yes. The engineer can push her own branch now. The correct review path is feature/ai-ml-v2 to integration/v2, and after all five roles are integrated and tested, integration/v2 to main. Use pull requests, not direct pushes to the shared branches.

Verified repository state

• main and integration/v2: commit fc0f269.

• feature/backend-v2: commit c98e9c2, five commits ahead of integration/v2.

• The AI/ML branch is not pushed yet; there were no open PRs when checked.

Official repository: https://github.com/rani400/Ai-powered-intelligent-sales-forecasting-platform-using-predictive-Analytics

Exact AI/ML ownership - only these files

File	Purpose
app/ml/__init__.py; preprocessing.py	Module and shared train/predict feature schema
app/ml/train.py; evaluate.py	Scikit-learn model training and saved metric reporting
app/ml/predict.py	Probability inference and labeled fallback
data/generate_demo_data.py	Deterministic synthetic-data generator
data/sample_training_data.csv	600 synthetic training observations
docs/ML_METHODOLOGY.md	Methodology, validation and known limitations
requirements.txt	APPEND only the ML packages listed below

Never include: .env or keys, .venv, generated .joblib/.pkl models, local DB files, frontend/, app/database.py, app/models/, app/routers/, app/services/, migrations/, or the QA test suite. Coordinate shared-file edits.

Accuracy & authorship: 600 synthetic rows; earlier model accuracy 64.67%, not a validated real-world forecast or 85%+ result. Integrating existing code is not the same as claiming original authorship.

1. Set the extracted ZIP location

After extracting the clean ZIP, change SRC below to the folder that actually contains app/ml/train.py. Verify both files exist before continuing.

```
set "SRC=%USERPROFILE%\Downloads\SalesGenie_AI_for_AIML\SalesGenie_AI"
dir "%SRC%\app\ml\train.py"
dir "%SRC%\data\sample_training_data.csv"
```

2. Clone team repo and make your own ML branch

```
cd /d "%USERPROFILE%\Downloads"
git clone
    https://github.com/rani400/Ai-powered-intelligent-sales-forecasting-platform-using-predictive-
Analytics.git
    salesgenie-aiml-work
cd /d "%USERPROFILE%\Downloads\salesgenie-aiml-work"
git fetch origin
git switch -c feature/ai-ml-v2 origin/integration/v2
git status
git config user.name
git config user.email
```

Make sure the commit identity belongs to the AI/ML engineer. She needs collaborator permission to push to the team repo; otherwise use a personal fork and a PR.

3. Commit A - synthetic data + feature preprocessing

```
if not exist app\ml mkdir app\ml
copy /Y "%SRC%\app\ml\__init__.py" "app\ml\__init__.py"
copy /Y "%SRC%\app\ml\preprocessing.py" "app\ml\preprocessing.py"
copy /Y "%SRC%\data\generate_demo_data.py" "data\generate_demo_data.py"
copy /Y "%SRC%\data\sample_training_data.csv" "data\sample_training_data.csv"
git add app/ml/__init__.py app/ml/preprocessing.py data/generate_demo_data.py
    data/sample_training_data.csv
git diff --cached --stat
git diff --cached --check
git commit -m "Integrate synthetic training data and ML preprocessing"
```

4. Commit B - training and evaluation

```
copy /Y "%SRC%\app\ml\train.py" "app\ml\train.py"
copy /Y "%SRC%\app\ml\evaluate.py" "app\ml\evaluate.py"
git add app/ml/train.py app/ml/evaluate.py
git diff --cached --check
git commit -m "Integrate conversion classifier training and evaluation"
```

5. Commit C - model prediction

```
copy /Y "%SRC%\app\ml\predict.py" "app\ml\predict.py"
git add app/ml/predict.py
git diff --cached --check
git commit -m "Integrate conversion prediction and labeled fallback"
```

6. Commit D - methodology and ML dependencies

```
copy /Y "%SRC%\docs\ML_METHODOLOGY.md" "docs\ML_METHODOLOGY.md"
notepad requirements.txt
```

Append only the following missing requirements, each once. Preserve every existing package:

```
pandas>=2.2,<3
numpy>=1.26,<3
scikit-learn>=1.5,<2
joblib>=1.4,<2
```

```
git add docs/ML_METHODOLOGY.md requirements.txt
git diff --cached --stat
git diff --cached --check
git commit -m "Document ML methodology and add model dependencies"
```

These commit titles say Integrate, not that one teammate originally wrote every line. Review each group. If a group is unchanged, skip the empty commit; do not manufacture history.

7. Push feature branch and open PR

```
git status
git log --oneline -5
git diff --name-status origin/integration/v2...HEAD
git push -u origin feature/ai-ml-v2
```

GitHub PR base: integration/v2 | Compare: feature/ai-ml-v2

Suggested PR title: Integrate AI/ML training, evaluation and prediction pipeline

Suggested PR description: Reproducible synthetic-data generation, feature preprocessing, logistic-regression model, held-out evaluation and inference with clearly labeled non-ML fallback. Model results are synthetic-only. End-to-end verification awaits backend and database integration.

Open as draft until backend and database dependencies are integrated. Do not merge straight to main.

8. Once database + backend PRs are in integration/v2

```
git fetch origin
git switch feature/ai-ml-v2
git merge origin/integration/v2
py -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.txt
.venv\Scripts\python.exe -m data.generate_demo_data
.venv\Scripts\python.exe -m app.ml.train
.venv\Scripts\python.exe -m pytest -q
git status
```

Resolve any actual requirements.txt conflict while keeping database, backend and ML dependencies. Regenerating data can modify the tracked CSV; commit it only if intended. Model .joblib artifacts and .env must remain untracked.

9. Final merge sequence

- AI/ML engineer pushes feature/ai-ml-v2 now; review via PR targeted to integration/v2.
- Database Engineer and Backend Developer merge compatible database/API changes to integration/v2.
- Merge reviewed AI/ML PR; Frontend and QA engineers then integrate/test their respective changes.
- Run migrations, all tests, and verify all four pages plus real Gemini behavior.
- Open a final reviewed PR from integration/v2 into main - only after the integrated app passes.